

# Oguzhan Gungordu

✉ [ogungordu3@gatech.edu](mailto:ogungordu3@gatech.edu) | +1 (470) 662-5973 | [in](#) | [Twitter](#) | [GitHub](#) | [Globe](#)

## EDUCATION

**Georgia Institute of Technology**, Atlanta, GA 2024 - Present  
**Ph.D.** in Machine Learning *GPA: 4.00/4.00*  
*Graduate Research Assistant; Advisor: Prof. Faramarz Fekri*  
*Research Topic: LLM Reasoning, LLM for Optimization, and Scientific Discovery with LLMs*

**Koc University**, Istanbul, Turkey 2019 - 2023  
**B.Sc.** in Electrical and Electronics Engineering *Area GPA: 3.99/4.00*  
*Thesis: Saliency-Aware End-to-End Learned Variable-Bitrate 360° Image Compression*  
**B.Sc.** in Physics *Area GPA: 4.00/4.00*  
*Thesis: Physics-Informed Deep Learning Solutions of Schrödinger Equation and Kronig-Penney Model*

## RESEARCH INTERESTS

Research interests lie in the areas of Machine Learning, Large Language Models, and Optimization. My current research focuses on LLM Reasoning (planning, self-improvement, and evidence-based reasoning with state-aware memory and retrieval-augmented generation) and LLMs for Optimization and Discovery (LLM-driven evolutionary frameworks for complex optimization and scientific problems).

## PUBLICATIONS

**O. Gungordu**, S. Xiong, and F. Fekri, "PathWise: Planning through World Model for Automated Heuristic Design via Self-Evolving LLMs," *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026. [📄](#)

**O. Gungordu** and A. M. Tekalp, "Saliency-Aware End-to-End Learned Variable-Bitrate 360-Degree Image Compression," *2024 IEEE International Conference on Image Processing (ICIP)*, Abu Dhabi, United Arab Emirates, 2024, pp. 1795-1801. [📄](#)

## PREPRINTS

**O. Gungordu**, A. Tursucular, S. Xiong, and F. Fekri, "EPOCH: Coverage-Aware Evolution with Experiential Preference Optimization for LLM-based Automated Heuristic Design," *Under review*, 2026.

**O. Gungordu**, S. Xiong, and F. Fekri, "MOSAIC: Adversarial Co-evolution of Specialist Heuristics and Problem Instances for LLM-based Automated Heuristic Design," *Under review*, 2026.

S. Xiong, A. Payani, **O. Gungordu**, and F. Fekri, "DIVE: Unlocking Self-Improvement in Frozen Language Models Through Diversity-Driven Skill Evolution," *Under review*, 2026.

S. Xiong, **O. Gungordu**, B. Johnson, J. C. Kerce, and F. Fekri, "Scaling Search-Augmented Reasoning Agents via Adaptive Information Control," *Workshop on Scaling Post-training for LLMs (SPOT), ICLR 2026*. [📄](#)

S. Xiong, **O. Gungordu**, B. Johnson, M. Okamoto, J. C. Kerce, and F. Fekri, "Deep-Verify: Evidence-Based Expert-Level Scientific Claim Verification," 2026. [📄](#)

## COURSEWORK

### Graduate

• Mathematical Foundations of Machine Learning (ECE7750) • Random Processes (ECE6601) • Deep Learning (CS7643) • Convex Optimization (ECE6270) • Graphical ML (ECE7751) • Computational Data Analysis (CSE6740) • Generative and Geometric Deep Learning (ECE8803)

### Undergraduate

Introduction to ML • Introduction to AI • Digital Signal Processing • Digital Image & Video Processing • Digital Speech & Audio Processing • Biomedical Signal Processing • Communication Networks • Analog & Digital Communication Laboratory

## RESEARCH EXPERIENCE

**Research Assistant, Koc University**  
Advised by Prof. Murat Tekalp for Bachelor Thesis

Jun. 2022 - Jun. 2024

- Conducting research primarily focused on omnidirectional image compression and saliency detection techniques to reduce requirements of VR headsets.
- Developed first “Saliency-aware End-to-end Learned Variable Rate 360° Image Compression” architecture, which is focused on the development of compression technique for 360° images according to saliency map, to reduce data size while preserving the most visually significant regions.

#### **Research Assistant, Koc University**

Jan. 2023 - Jul. 2023

Advised by Assoc. Prof. Menderes Işkın for Bachelor Thesis

- Researched possibilities of solving physical models using computer vision techniques, especially for predicting Schrödinger equation solutions and energy band-gap of semiconductors.
- Implemented a deep learning architecture that solves the Schrödinger equation and predicts ground-state energy in 2D mesh utilizing CNN. [\[Report\]](#)
- Implemented a deep learning architecture that accurately predicts energy bands of periodic potentials in the Kronig-Penney model of semiconductors. [\[Report\]](#)

### **WORK EXPERIENCE**

#### **MLOps Engineer, CareX AI**

Oct. 2021 - Aug. 2024

Palo Alto, California/USA (Remote)

- CareX AI delivers a video-based remote patient monitoring platform that transforms any camera-equipped device into a remote patient monitoring solution.
- Developing vitals (heart rate, heart rate variability, SPO2 level, respiration rate, blood pressure, blood sugar, etc.) extraction methods from face and fingertip videos using biomedical signal processing, machine learning, and computer vision techniques, especially face recognition and denoising autoencoders.
- Implemented a transformer-based video motion magnification technique to track respiration rate by observing chest motion with optical flow and designed a lightweight model for real-time processing on edge devices.
- Optimized data acquisition process and deployment CI/CD pipeline to simplify PPG extraction from fingertip and face videos by using Docker and AWS.

#### **Access Network Engineer Intern, Turkcell**

Aug. 2021 - Sep. 2021

Istanbul/Turkey

- Experienced installation and implementation of the network design and radio transmission.
- Created the topology of the Microwave Radio Links and fiber lines to prevent signal interference in local areas and optimized the network map of routers and switches.

#### **Software Engineering Intern, Türkiye İŞ Bankası**

Jun. 2021 - Jul. 2021

Istanbul/Turkey

- Worked on the bank’s AI platform, contributing to the management of model storage accessibility.

### **AWARDS & HONORS**

#### **Georgia Tech ECE Departmental Fellowship**

2024

#### **Valedictorian of the Department of Physics**

2023

#### **College of Science Top Ranked Student (3<sup>rd</sup> Place)**

2023

#### **B.Sc. in Electrical and Electronics with *Summa Cum Laude***

2023

#### **TUBITAK Undergraduate Scholarship**

2020 - 2023

- This scholarship is given to students who sustain academic excellence by The Scientific and Technological Research Council of Turkey (TUBITAK).

#### **Vehbi Koc Academic Honor Award**

2018 - 2023

#### **Full Merit Undergraduate Scholarship**

2018 - 2023

- Ranked 115th among approximately 2.4 million candidates in the nationwide university entrance exam.

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|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TEACHING</b>             | <b>Graduate Teaching Assistant, <i>Introduction to Signal Processing</i></b><br>ECE 2026, Georgia Institute of Technology <span style="float: right;">Fall'24, Spring'25</span><br>Hosted office hours, graded homework assignments, and conducted lab sessions to support students in the course.                                                                                                                                                                                                     |
|                             | <b>Undergraduate Teaching Assistant, <i>Differential Equations</i></b><br>MATH 204, Koc University <span style="float: right;">Spring'22</span><br>Held office hours to assist students with questions and provide support for the course.                                                                                                                                                                                                                                                             |
|                             | <b>Learning and Teaching Center Tutor, <i>General Physics II</i></b><br>PHYS 102, Koc University <span style="float: right;">Fall'20</span><br>Served as a tutor, assisting students in improving their understanding.                                                                                                                                                                                                                                                                                 |
| <b>INDEPENDENT PROJECTS</b> | <b>Speech Coding: The Method of Parametric Coder and Linear Predictive Coding (LPC) Vocoder</b><br>Digital Speech & Audio Processing Final Project Advised by Prof. Engin Erzin <ul style="list-style-type: none"> <li>• Comparison of Parametric Coder and LPC Vocoder are discussed in terms of perceptual quality and SSNR. Also, the performance of the different CODEC that represents bit allocations is analyzed in detail in the case of Parametric Coder. <a href="#">[Report]</a></li> </ul> |
|                             | <b>Heart Rate Monitoring Using Seismocardiogram Signal: The Method of Template SCG</b><br>Biomedical Signal Processing Final Project Advised by Assoc. Prof. Beren Semiz <ul style="list-style-type: none"> <li>• Researched measuring the heart rate and heart rate variability in the long-term without requiring an ECG measurement. We demonstrate that the SCG signal is sufficient to get information about cardiac mechanics. <a href="#">[Report]</a></li> </ul>                               |
| <b>SKILLS</b>               | <b>Languages:</b> English (Fluent, TOEFL IBT 102/120), Turkish (Native)<br><b>Programming:</b> Python (PyTorch, TensorFlow, OpenCV), FFmpeg, MATLAB, AWS, Docker, SQL                                                                                                                                                                                                                                                                                                                                  |
| <b>VOLUNTEER EXPERIENCE</b> | <b>Global Volunteer, AIESEC</b> <span style="float: right;">2019</span> <ul style="list-style-type: none"> <li>• Over an 8-week period, an orphanage in Cairo, Egypt, was supported through my volunteer work as part of the “Social Project in Egyptian Society for Integrated Development.” A key focus was placed on helping underprivileged children to improve their computer skills.</li> </ul>                                                                                                  |
|                             | <b>Student Mentor, Newcomer Advising Center, Koc University</b> <span style="float: right;">2020 - 2022</span> <ul style="list-style-type: none"> <li>• Assisted newcomer students with their social and academic adaptation by organizing biweekly meetings and events.</li> </ul>                                                                                                                                                                                                                    |